

Chapter 6.

Counting

計數

6.1 The Basics of Counting

6.2 The Pigeonhole Principle 鴿籠原理

6.3 Permutations and Combinations

6.4 Binomial Coefficients and Identities

6.5 Generalized Permutations and Combinations

Basic Counting Principles:

The Product Rule

乘法規則

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

Example: How many bit strings of length seven are there?

0, 1

$\underbrace{1 \quad 1 \quad 1 \quad \dots \quad 1}_{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2} = 2^7 = 128$

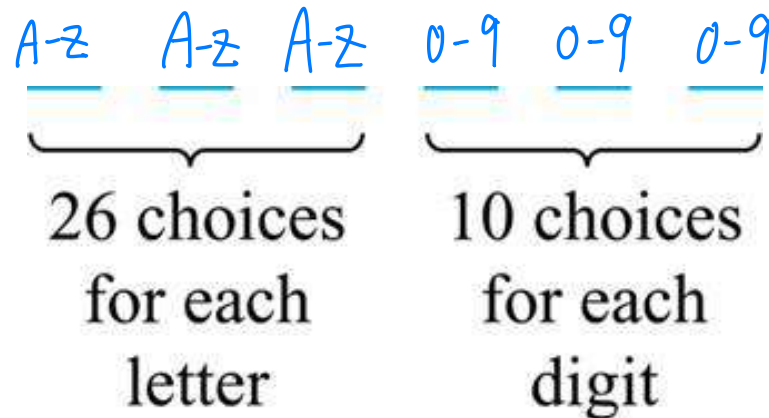
Solution: Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.

The Product Rule

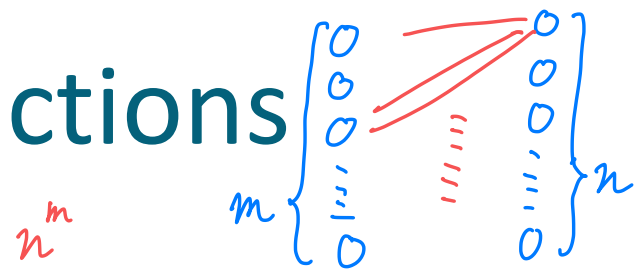
Example: How many **different license plates** can be made if each plate contains a sequence of **three uppercase English letters followed by three digits**?

Solution: By the product rule,

there are $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$ different possible license plates.



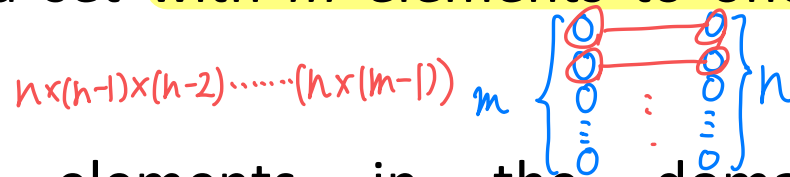
Counting Functions



Counting Functions: **How many functions** are there from a set with m elements to a set with n elements?

Solution: Since a function represents a choice of one of the n elements of the codomain for each of the m elements in the domain, the product rule tells us that there are $n \cdot n \cdots n = n^m$ such functions.

Counting One-to-One Functions: **How many one-to-one functions** are there from a set with m elements to one with n elements?



Solution: Suppose the elements in the domain are $a_1, a_2, \dots, a_m$. There are n ways to choose the value of a_1 and $n-1$ ways to choose a_2 , etc. The product rule tells us that there are $n(n-1)(n-2)\cdots(n-m+1)$ such functions.

Telephone Numbering Plan

Example: The *North American numbering plan (NANP)* specifies that a telephone number consists of 10 digits, consisting of a **three-digit area code**, a **three-digit office code**, and a **four-digit station code**. There are some restrictions on the digits.

- Let X denote a digit from 0 through 9.
- Let N denote a digit from 2 through 9.
- Let Y denote a digit that is 0 or 1.
- In the **old plan** (in use in the 1960s) the format was $NYX-NXX-XXXX$. $8 \times 2 \times 10 \times 8^2 \times 10 \times 10^4$
- In the **new plan**, the format is $NXX-NXX-XXXX$.

How many different telephone numbers are **possible under the old plan and the new plan?**

Solution: Use the Product Rule.

- There are $8 \cdot 2 \cdot 10 = 160$ area codes with the format NYX .
- There are $8 \cdot 10 \cdot 10 = 800$ area codes with the format NXX .
- There are $8 \cdot 8 \cdot 10 = 640$ office codes with the format NNX .
- There are $10 \cdot 10 \cdot 10 \cdot 10 = 10,000$ station codes with the format $XXXX$.

Number of old plan telephone numbers: $160 \cdot 640 \cdot 10,000 = 1,024,000,000$.

Number of new plan telephone numbers: $800 \cdot 800 \cdot 10,000 = 6,400,000,000$.

Counting Subsets of a Finite Set

Counting Subsets of a Finite Set: Use the product rule to show that the number of **different subsets** of a **finite set** S is $2^{|S|}$. (In Section 5.1, mathematical induction was used to prove this same result.)

(Handwritten note: 集合的个数 points to $|S|$)

Solution: When the elements of S are listed in an arbitrary order, there is a one-to-one correspondence between subsets of S and bit strings of length $|S|$. When the i th element is in the subset, the bit string has a 1 in the i th position and a 0 otherwise.

By the product rule, there are $2^{|S|}$ such bit strings, and therefore $2^{|S|}$ subsets.

Product Rule in Terms of Sets

If $A_1, A_2, \dots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements of each set.

The task of choosing an element in the **Cartesian product** (笛卡兒積) $A_1 \times A_2 \times \dots \times A_m$ is done by choosing an element in A_1 , an element in A_2 , ..., and an element in A_m .

$$A_1 \times A_2 = \{(a,0), (a,1), (b,0), (b,1), (c,0), (c,1)\} \quad A_1 = \{a, b, c\}$$

By the product rule, it follows that: $A_2 = \{0, 1\}$

$$|A_1 \times A_2 \times \dots \times A_m| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_m|$$

DNA and Genomes

A *gene* is a segment of a DNA molecule that encodes a particular protein and the entirety of genetic information of an organism is called its *genome*.

DNA molecules consist of two strands of blocks known as nucleotides. Each nucleotide is composed of bases: adenine (A), cytosine (C), guanine (G), or thymine (T).

The DNA of bacteria has between 10^5 and 10^7 links (one of the four bases). Mammals have between 10^8 and 10^{10} links. So, by the product rule there are at least 4^{10^5} different sequences of bases in the DNA of bacteria and 4^{10^8} different sequences of bases in the DNA of mammals. The human genome includes approximately 23,000 genes, each with 1,000 or more links.

Biologists, mathematicians, and computer scientists all work on determining the DNA sequence (genome) of different organisms.

Basic Counting Principles:

The Sum Rule

The Sum Rule: If a task can be done **either in one of n_1 ways or in one of n_2** , where **none of the set of n_1 ways is the same as any of the n_2 ways**, then **there are $n_1 + n_2$ ways to do the task.**

Example: The mathematics department must choose **either a student or a faculty member** as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and **no one is both a faculty member and a student.**

Solution: By the sum rule it follows that there are **$37 + 83 = 120$** possible ways to pick a representative.

The Sum Rule in terms of sets.

The sum rule can be phrased in terms of sets.

$|A \cup B| = |A| + |B|$ as long as A and B are disjoint sets.

Or more generally,

$$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

when $A_i \cap A_j = \emptyset$ for all i, j .

The case where the sets have elements in common will be discussed when we consider the subtraction rule and taken up fully in Chapter 8.

Combining the Sum and Product Rule

Example: Suppose statement labels in a programming language can be either **a single letter** or **a letter followed by a digit**. Find the number of possible labels.

Solution: Use the product rule.

$$26 + 26 \cdot 10 = 286$$

$$(\underline{A-Z}) + (\underline{A-Z} \ \underline{0-9})$$

Counting Passwords

Combining the sum and product rule allows us to solve more complex problems.

Example: Each user on a computer system has a password, which is **six to eight characters long**, where each character is an **uppercase letter or a digit**. Each password must **contain at least one digit**. How many possible passwords are there?

全情况 - 只有英文的

Solution: Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8.

全: $\frac{0}{36} \frac{A}{36} \frac{0}{36} \frac{A}{36} \dots \frac{0}{36} \frac{A}{36}$ 只有英文: $\frac{A}{26} \frac{A}{26} \dots \frac{A}{26}$

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6 , P_7 , and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:

$$P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560.$$

$$P_7 = 36^7 - 26^7 =$$

$$78,364,164,096 - 8,031,810,176 = 70,332,353,920.$$

$$P_8 = 36^8 - 26^8 =$$

$$2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880.$$

Consequently, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

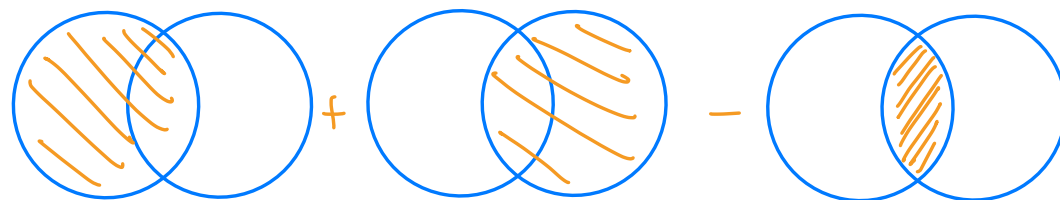
Basic Counting Principles:

Subtraction Rule

Subtraction Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ **minus the number of ways to do the task that are common** to the two different ways.

Also known as, the *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

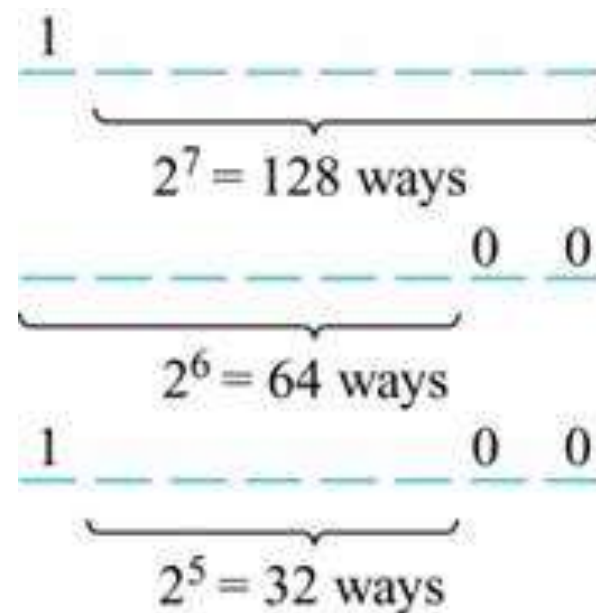


Counting Bit Strings

Example: How many bit strings of length eight **either start with a 1 bit or end with the two bits 00?**

Solution: Use the subtraction rule.

- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that end with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$



Hence, the number is $128 + 64 - 32 = 160$.

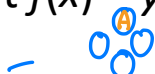
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Basic Counting Principles: Division Rule

Division Rule: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

Restated in terms of sets: If the finite set A is the union of n pairwise disjoint subsets each with d elements, then $n = |A|/d$.

In terms of functions: If f is a function from A to B , where both are finite sets, and for every value $y \in B$ there are exactly d values $x \in A$ such that $f(x) = y$, then $|B| = |A|/d$.

Example: How many ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left and right neighbor?
 → 4張椅, 鎖定A, 椅子挑剩下的人 $3 \times 2 \times 1 = 6$

Solution: Number the seats around the table from 1 to 4 proceeding clockwise. There are four ways to select the person for seat 1, 3 for seat 2, 2 for seat 3, and one way for seat 4. Thus there are $4! = 24$ ways to order the four people. But since two seatings are the same when each person has the same left and right neighbor, for every choice for seat 1, we get the same seating.

Therefore, by the division rule, **there are $24/4 = 6$ different seating arrangements.**

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6.2 The Pigeonhole Principle

6.3 Permutations and Combinations

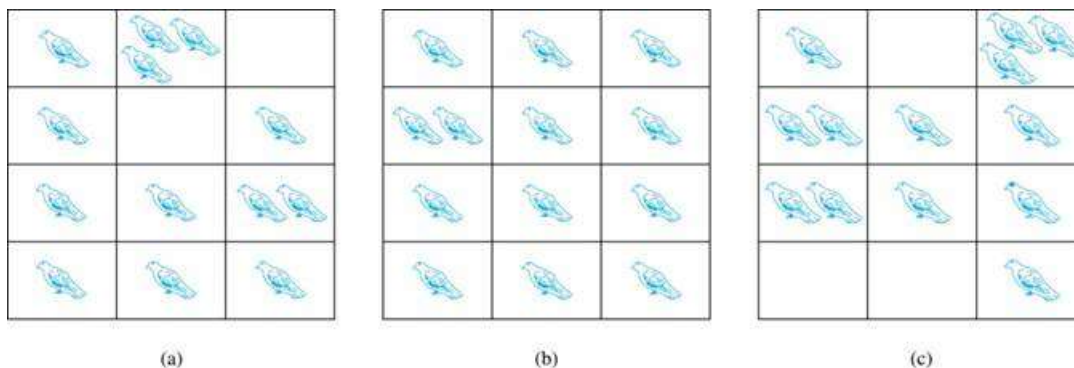
6.4 Binomial Coefficients and Identities

6.5 Generalized Permutations and Combinations

The Pigeonhole Principle₁

(鴿籠原理)

If a flock of 13 pigeons roosts in a set of 12 pigeonholes, one of the pigeonholes must have more than 1 pigeon.



Pigeonhole Principle: If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Proof: We use a proof by contraposition. Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k . This contradicts the statement that we have $k + 1$ objects.

The Pigeonhole Principle₂

Corollary 1: A function f from a set with $k + 1$ elements to a set with k elements is not one-to-one.

Proof: Use the pigeonhole principle. $k+1$ $\left\{ \begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \end{array} \right\}$ $\left\{ \begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \end{array} \right\}^k$ Not 1-1

- Create a box for each element y in the codomain of f .
- Put in the box for y all of the elements x from the domain such that $f(x) = y$.
- Because there are $k + 1$ elements and only k boxes, at least one box has two or more elements.

Hence, f can't be one-to-one.

Pigeonhole Principle

Example: Among any group of 367 people, **there must be at least two with the same birthday**, because there are only 366 possible birthdays.

The Generalized Pigeonhole Principle₁

一般形式

The Generalized Pigeonhole Principle: If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Proof: We use a proof by contraposition. Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects is at most

鴿籠原理
↓
極端情況

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N,$$

where the inequality $\lceil N/k \rceil < \lceil N/k \rceil + 1$ has been used. This is a contradiction because there are a total of n objects.

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

每月先允許放8個人
那剩下還有4人必會被隨機分到其他月

The Generalized Pigeonhole Principle₂

Example: a) How many cards must be selected from a standard deck of **52 cards** to guarantee that **at least three cards** of the same suit (花色) are chosen?

$$\begin{array}{c} \circ \quad \circ \quad \circ \quad \circ \\ \hline A \quad B \quad C \quad D \end{array} \quad 8+1=9$$

b) How many must be selected to guarantee that **at least three hearts** are selected?

Solution: a) We assume four boxes; one for each suit. Using the generalized pigeonhole principle, at least one box contains at least $\lceil N/4 \rceil$ cards. At least three cards of one suit are selected if $\lceil N/4 \rceil \geq 3$. The smallest integer N such that $\lceil N/4 \rceil \geq 3$ is $N = 2 \cdot 4 + 1 = 9$.

b) A deck contains 13 hearts and 39 cards which are not hearts. So, if we select 41 cards, we may have 39 cards which are not hearts along with 2 hearts. However, when we select **42 cards**, we must have at least three hearts. (Note that the generalized pigeonhole principle is not used here.)

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Permutations

(排列、交換)

Definition: A *permutation* of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of r elements of a set is called an *r -permutation*.

Example: Let $S = \{1, 2, 3\}$.

- The ordered arrangement 3,1,2 is a permutation of S .
- The ordered arrangement 3,2 is a 2-permutation of S .

The number of r -permutations of a set with n elements is denoted by $P(n, r)$. P_r^n

- The 2-permutations of $S = \{1, 2, 3\}$ are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence, $P(3, 2) = 6$.

A Formula for the Number of Permutations

Theorem 1: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

Proof: Use the product rule. The first element can be chosen in n ways. The second in $n - 1$ ways, and so on until there are $(n - (r - 1))$ ways to choose the last element.

Note that $P(n, 0) = 1$, since there is only one way to order zero elements.

Corollary 1: If n and r are integers with $1 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n - r)!}$$

25 Solving Counting Problems by Counting Permutations₁

Example: How many ways are there to select a first-prize winner, a second prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution:

$$\begin{array}{ccc} \underline{100} & \underline{99} & \underline{98} \\ \hookrightarrow & 100 \times 99 \times 98 & \end{array}$$

$$P(100,3) = 100 \cdot 99 \cdot 98 = 970,200$$

26 Solving Counting Problems by Counting Permutations₂

Example: Suppose that a saleswoman has to visit **eight different cities**. She must begin her trip in a specified city, but **she can visit the other seven cities in any order** she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution: The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

27 Solving Counting Problems by Counting Permutations₃

Example: How many permutations of the letters *ABCDEFGH* **contain the string *ABC***?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

$$\boxed{ABC}DEFGH = 6! = 720$$

$$\underline{6} \underline{5} \underline{4} \underline{3} \underline{2} \underline{1} = 6!$$

Combinations₁

(組合)

Definition: An **r -combination** of elements of a set is an unordered selection of r elements from the set. Thus, an r -combination is simply a subset of the set with r elements. The number of r -combinations of a set with n distinct elements is denoted by $C(n, r)$. The notation $\binom{n}{r}$ is also used and is called a **binomial coefficient (二項式係數)**. (We will see the notation again in the binomial theorem in Section 6.4.)

Example: Let S be the set $\{a, b, c, d\}$. Then $\{a, c, d\}$ is a 3-combination from S . It is the same as $\{d, c, a\}$ since the order listed does not matter.

$$C_2^4 = \frac{4 \times 3}{2 \times 1} = 6$$

$C(4, 2) = 6$ because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Combinations₂

Theorem 2: The number of r -combinations of a set with n elements, where $n \geq r \geq 0$, equals

<Review>

$$C_{10}^{10} = C_2^{10}$$

$$C_k^n = C_p^n, \text{ if } k+p=n$$

$$C(n, r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n, r) \cdot P(r, r)$.
Therefore

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n! / (n-r)!}{r! / (r-r)!} = \frac{n!}{(n-r)!r!}.$$

Combinations₃

Example: How many poker hands of **five cards** can be dealt from a **standard deck of 52 cards**? Also, how many ways are there to **select 47 cards** from a deck of 52 cards?

C_{52}^{52} 捨棄47張
 C_{47}^{52} 捨棄5張

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$C(52, 5) = \frac{52!}{5!47!}$$

$$= \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2,598,960$$

$C_{52}^{52n} = C_{47}^{52n}$ $k+p=n$

The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2,598,960$$

Combinations₄

Corollary 2: Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n - r)$.

Proof: From Theorem 2, it follows that

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

and

$$C(n, n-r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!}.$$

Hence, $C(n, r) = C(n, n - r)$.

Combinatorial Proofs₁

Definition 1: A *combinatorial proof* of an identity is a proof that uses one of the following methods.

- A *double counting proof* uses counting arguments to prove that both sides of an identity count the same objects, but in different ways.
- A *bijective proof* shows that there is a bijection between the sets of objects counted by the two sides of the identity.

Combinatorial Proofs₂

Here are two combinatorial proofs that

$$C(n, r) = C(n, n - r)$$

when r and n are nonnegative integers with $r < n$:

- ***Bijjective Proof:*** Suppose that S is a set with n elements. The function that maps a subset A of S to $\bar{A}$ is a bijection between **the subsets of S with r elements and the subsets with $n - r$ elements**. Since there is a bijection between the two sets, they must have the same number of elements.
- ***Double Counting Proof:*** By definition the number of subsets of S with r elements is $C(n, r)$. **Each subset A of S can also be described by specifying which elements are not in A** , i.e., those which are in $\bar{A}$. Since the complement of a subset of S with r elements has $n - r$ elements, there are also $C(n, n - r)$ subsets of S with r elements.

Combinations₅

Example: How many ways are there to **select five players** from a **10-member** tennis team to make a trip to a match at another school.

$$C_{5}^{10} = \frac{10!}{5!5!} = 252$$

Solution: By Theorem 2, the number of combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

$$C_6^{30} = \frac{30 \times 29 \times 28 \times 27 \times 26 \times 25}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

Example: A group of **30 people** have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of **six people** to go on this mission?

Solution: By Theorem 2, the number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775$$

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6.4 Binomial Coefficients and Identities

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Powers of Binomial Expressions

Definition: A **binomial expression** is the sum of two terms, such as $x + y$. (More generally, these terms can be products of constants and variables.)

- We can use counting principles to find the coefficients in the expansion of $(x + y)^n$ where n is a positive integer.
- To illustrate this idea, we first look at the process of expanding $(x + y)^3$.
- $(x + y)(x + y)(x + y)$ expands into a sum of terms that are the product of a term from each of the three sums.
- Terms of the form x^3 , x^2y , xy^2 , y^3 arise. The question is what are the coefficients?
 - To obtain x^3 , an x must be chosen from each of the sums. There is only one way to do this. So, the coefficient of x^3 is 1.
 - To obtain x^2y , an x must be chosen from two of the sums and a y from the other. There are $\binom{3}{2}$ ways to do this and so the coefficient of x^2y is 3.
- To obtain xy^2 , an x must be chosen from one of the sums and a y from the other two. There are $\binom{3}{1}$ ways to do this and so the coefficient of xy^2 is 3.
- To obtain y^3 , a y must be chosen from each of the sums. There is only one way to do this. So, the coefficient of y^3 is 1.

We have used a counting argument to show that $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$. Next we present the binomial theorem gives the coefficients of the terms in the expansion of $(x + y)^n$.

Binomial Theorem

二項式定理 $\rightarrow 2^n = C_0^n + C_1^n + C_2^n + \dots + C_n^n$

Binomial Theorem: Let x and y be variables, and n a nonnegative integer. Then: $2^4 = C_0^4 + C_1^4 + C_2^4 + C_3^4 + C_4^4$

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \dots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n.$$

Proof: We use combinatorial reasoning . The terms in the expansion of $(x + y)^n$ are of the form $x^{n-j} y^j$ for $j = 0, 1, 2, \dots, n$. To form the term $x^{n-j} y^j$, it is necessary to choose $n-j$ x s from the n sums. Therefore, the coefficient of $x^{n-j} y^j$ is $\binom{n}{n-j}$ which equals $\binom{n}{j}$.

Using the Binomial Theorem

Example: What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$?

$$\hookrightarrow (2x + (-3y))^{25} = \binom{25}{12} (2x)^{12} + (-3y)^{13} \Rightarrow \binom{25}{12} \times 2^{12} \times (-3)^{13}$$

find coefficients:

Solution: We view the expression as $(2x + (-3y))^{25}$.
By the binomial theorem

$$(2x + (-3y))^{25} = \sum_{j=0}^{25} \binom{25}{j} (2x)^{25-j} (-3y)^j$$

Consequently, the coefficient of $x^{12}y^{13}$ in the expansion is obtained when $j = 13$.

$$\binom{25}{13} 2^{12} (-3)^{13} = -\frac{25!}{13! 12!} 2^{12} 3^{13}.$$

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x+y)^3 = \boxed{}x^3 + \boxed{?}x^2y + \boxed{}xy^2 + \boxed{}y^3$$

Assume that coefficient is unknown
find x^2y coefficients?

$$(x+y)^3 = (x+y)(x+y)(x+y) = \dots \text{展开}$$

or 三个位子, 两个放 x , 一个放 $y \Rightarrow C_2^3 (C_1^3)$

$x \cdot y$ $x \cdot y$ $x \cdot y$

$$(x+y)^3 = \boxed{}x^3 + \boxed{?}x^2y + \boxed{}xy^2 + \boxed{}y^3$$

以 x
角度

$$C_3^3$$

$$C_2^3$$

$$C_1^3$$

$$C_0^3$$

A Useful Identity

Corollary 1: With $n \geq 0$, $\sum_{k=0}^n \binom{n}{k} = 2^n$. $2^n = C_0^n + C_1^n + \dots + C_n^n$

Proof (using binomial theorem): With $x = 1$ and $y = 1$, from the binomial theorem we see that:

$$2^n = (1+1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{(n-k)} = \sum_{k=0}^n \binom{n}{k}.$$

Proof (combinatorial): Consider the subsets of a set with n elements. There are $\binom{n}{0}$ subsets with zero elements, $\binom{n}{1}$ with one element, $\binom{n}{2}$ with two elements, ..., and $\binom{n}{n}$ with n elements. Therefore the total is $\sum_{k=0}^n \binom{n}{k}$.

Since, we know that a set with n elements has 2^n subsets, we conclude: $\sum_{k=0}^n \binom{n}{k} = 2^n$.

Pascal's Identity



Blaise Pascal
(1623-1662)

Pascal's Identity: If n and k are integers with $n \geq k \geq 0$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

假设一个班31个人
挑5个人:
- 有挑到你: C_4^{30}
- 没挑到你: C_5^{30}
 $C_5^{31} = C_4^{30} + C_5^{30}$

Proof (combinatorial): Let T be a set where $|T| = n + 1$, $a \in T$, and $S = T - \{a\}$. There are $\binom{n+1}{k}$ subsets of T containing k elements.

Each of these subsets either:

- contains a with $k - 1$ other elements, or
- contains k elements of S and not a .

There are

- $\binom{n}{k-1}$ subsets of k elements that contain a , since there are $\binom{n}{k-1}$ subsets of $k - 1$ elements of S ,
- $\binom{n}{k}$ subsets of k elements of T that do not contain a , because there are $\binom{n}{k}$ subsets of k elements of S .

Hence,
$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

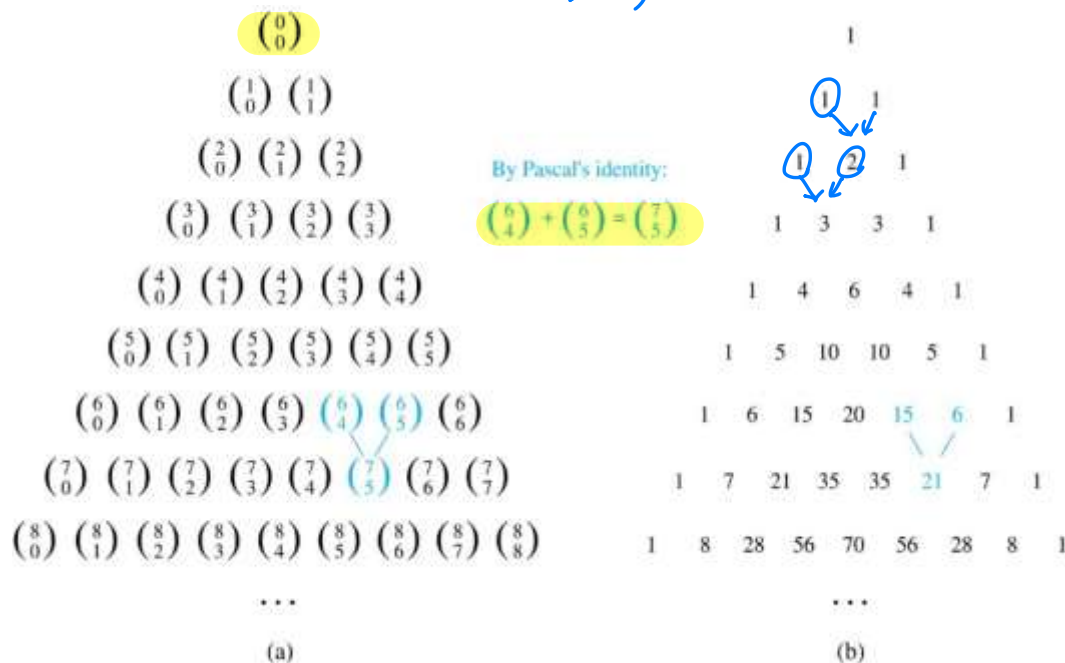
See Exercise 19 for
an algebraic proof.

Pascal's Triangle

$$C_0^0 = 1 \left(\frac{0!}{0!0!} = 1 \right)$$

The n th row in the triangle consists of the binomial coefficients

$$\binom{n}{k}, \quad k = 0, 1, \dots, n.$$



By Pascal's identity, adding two adjacent binomial coefficients results in the binomial coefficient in the next row between these two coefficients.

[Jump to long description](#)

Chapter 6.

Counting

6.1 The Basics of Counting

6.2 The Pigeonhole Principle

6.3 Permutations and Combinations P.C

6.4 Binomial Coefficients and Identities

6.5 Generalized Permutations and Combinations H

Permutations with Repetition (重複排列)

Theorem 1: The number of r -permutations of a set of n objects **with repetition** allowed is n^r . $\underbrace{\underbrace{n \quad n \quad n \quad n \quad \dots \quad n}_r}_{\Rightarrow n^r}$

Proof: There are n ways to select an element of the set for each of the r positions in the r -permutation when repetition is allowed. Hence, by the product rule there are n^r r -permutations with repetition.

Example: How many strings of **length** r can be formed from the uppercase letters of the English alphabet?

Solution: The number of such strings is 26^r , which is the number of r -permutations of a set with 26 elements.

Combinations with Repetition¹

(重複組合)

Example: How many ways are there to **select five bills** from a box containing at least five of each of the following denominations: \$1, \$2, \$5, \$10, \$20, \$50, and \$100?

Solution: Place the selected bills in the appropriate position of a cash box illustrated below:

未知數 (a~g) 個數

$$a + b + c + d + e + f + g = 5$$

How many way?

等於多少?

$$a + b + c + d + e + f + g + h = 5$$

上下相加 -1

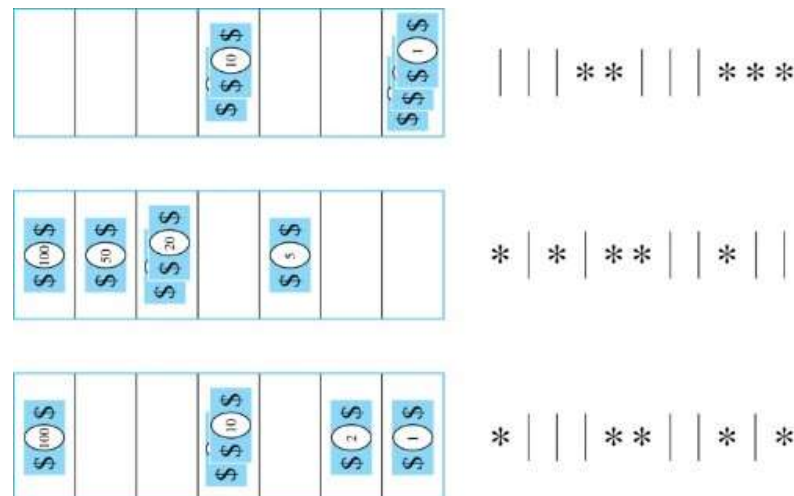
$$H_5^7 = C_5^{11}$$

下面多少寫多少

Combinations with Repetition₂

Some possible ways of placing the five bills:

The number of ways to select five bills corresponds to the number of ways to arrange six bars and five stars in a row.



This is the number of unordered selections of 5 objects from a set of 11. Hence, there are

$$C(11, 5) = \frac{11!}{5!6!} = 462$$

ways to choose five bills with seven types of bills.

$$\textcircled{1} \quad x_1 + x_2 + x_3 = 11$$

$$x_1, x_2, x_3 \geq 0$$

$$H_{11}^3 = c_{11}^{13} = c_{12}^{13}$$

$$\textcircled{2} \quad x_1, x_2, x_3 = 11$$

$$x_1, x_2, x_3 \geq 1$$

$$\frac{x_1}{0} \quad \frac{x_2}{0} \quad \frac{x_3}{0} \quad H_{\frac{0}{0}}^3 = c_{\frac{0}{0}}^{10} = c_{\frac{0}{2}}^{10}$$

Combinations with Repetition₃

Theorem 2: The number of r -combinations from a set with n elements when repetition of elements is allowed is

$$C(n + r - 1, r) = C(n + r - 1, n - 1)$$

Proof: Each r -combination of a set with n elements with repetition allowed can be represented by a list of $n - 1$ bars and r stars. The bars mark the n cells containing a star for each time the i th element of the set occurs in the combination.

The number of such lists is $C(n + r - 1, r)$, because each list is a choice of the r positions to place the stars, from the total of $n + r - 1$ positions to place the stars and the bars. This is also equal to $C(n + r - 1, n - 1)$, which is the number of ways to place the $n - 1$ bars.

Combinations with Repetition₄

Example: How many solutions does the equation

$$\overset{a}{x_1} + \overset{b}{x_2} + \overset{c}{x_3} = 11 \quad H_{11}^3 = C_{11}^{13}$$

have, where x_1 , x_2 and x_3 are **nonnegative integers**?

Solution: Each solution corresponds to a way to select 11 items from a set with three elements; x_1 elements of type one, x_2 of type two, and x_3 of type three.

$$H_{11}^3 = C_{11}^{13}$$

By Theorem 2 it follows that there are

$$C(3+11-1, 11) = C(13, 11) = C(13, 2) = \frac{13 \cdot 2}{1 \cdot 2} = 78$$

solutions.

Combinations with Repetition₅



Example: Suppose that a cookie shop has **four different kinds of cookies**. How many different ways can **six cookies be chosen**?

Solution: The number of ways to choose six cookies is the number of 6-combinations of a set with four elements. By Theorem 2

$$C(9, 6) = C(9, 3) = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$$

$$a + b + c + d = 6$$

$$H_6^4 = C_6^9$$

is the number of ways to choose six cookies from the four kinds.

Summarizing the Formulas for Counting Permutations and Combinations with and without Repetition

TABLE 1 Combinations and Permutations With and Without Repetition.		
Type	Repetition Allowed?	Formula
r -permutations	No	$\frac{n!}{(n-r)!}$
r -combinations	No	$\frac{n!}{r!(n-r)!}$
r -permutations	Yes	n^r
r -combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$

50 Permutations with Indistinguishable Objects

$\{S_1, S_2, S_3, U, E, C_1, C_2\}$

Example: How many different strings can be made by reordering the letters of the word **SUCCESS**.

Solution: There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!}$$

$\frac{7!}{3!2!1!1!}$

The reasoning can be generalized to the following theorem. →

Permutations with Indistinguishable Objects₂

Theorem 3: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, ..., and n_k indistinguishable objects of type k , is:

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Proof: By the product rule the total number of permutations is:

$C(n, n_1) C(n - n_1, n_2) \cdots C(n - n_1 - n_2 - \cdots - n_k, n_k)$ since:

- The n_1 objects of type one can be placed in the n positions in $C(n, n_1)$ ways, leaving $n - n_1$ positions.
- Then the n_2 objects of type two can be placed in the $n - n_1$ positions in $C(n - n_1, n_2)$ ways, leaving $n - n_1 - n_2$ positions.
- Continue in this fashion, until n_k objects of type k are placed in $C(n - n_1 - n_2 - \cdots - n_k, n_k)$ ways.

The product can be manipulated into the desired result as follows:

$$\frac{n!}{n_1!(n - n_1)!} \frac{(n - n_1)!}{n_2!(n - n_1 - n_2)!} \cdots \frac{(n - n_1 - \cdots - n_{k-1})!}{n_k!0!} = \frac{n!}{n_1!n_2!\cdots n_k!}.$$

Distributing Objects into Boxes₁

Many counting problems can be solved by counting the ways objects can be placed in boxes.

- The **objects** may be either different from each other (*distinguishable*) or identical (*indistinguishable*).
- The **boxes** may be labeled (*distinguishable*) or unlabeled (*indistinguishable*).

Distributing Objects into Boxes₂

Distinguishable objects and distinguishable boxes.

- Example: There are $52!/(5!5!5!5!32!)$ ways to distribute hands of **5 cards each to four players.**
- There are $n!/(n_1!n_2! \cdots n_k!)$ ways to distribute n distinguishable objects into k distinguishable boxes.

Indistinguishable objects and distinguishable boxes.

- Example: There are $C(8 + 10 - 1, 10) = C(17,10) = 19,448$ ways ($H_{10}^8 = C_{10}^{17}$) to place **10 indistinguishable objects into 8 distinguishable boxes.**
- There are $C(n + r - 1, n - 1)$ ways to place r indistinguishable objects into n distinguishable boxes.

Distributing Objects into Boxes₃

$$a+b+c+d=3 \times \text{不可用公式}$$

Distinguishable objects and indistinguishable boxes.

- Example: There are 14 ways to put **four employees into three indistinguishable offices** (see Example 10).
- **No simple formula**

Indistinguishable objects and indistinguishable boxes.

- Example: There are 9 ways to pack **six copies of the same book into four identical boxes** (see Example 11).
- **No simple formula**

6,
5, 1
4, 2
4, 1, 1
3, 3
3, 2, 1
3, 1, 1, 1
2, 2, 2
2, 2, 1, 1